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Competition, Product Differentiation and Crises: Evidence from 18 Million Securitized Loans

Contents

1	Big picture	2
2	Data and measurement	2
2.1	Loan-level data	2
2.2	Product differentiation measure	3
2.3	Sponsor data and parent identification	3
2.4	Local sponsor competition	3
2.5	County-level controls	3
3	Theory and empirical specifications	4
3.1	Theory: product differentiation as competition escape	4
3.2	Borrower types and competition thresholds	4
3.3	Baseline loan-level specification	4
3.4	Within-originator specification	5
3.5	HUD-rule dynamic difference-in-differences	5
3.6	Marginal borrower spline specification	5
3.7	Delinquency analysis	5
4	Identification and empirical design	6
4.1	What identifies the baseline relation	6
4.2	Why within-originator tests matter	6
4.3	Why the HUD rule is useful	6
4.4	Remaining limitations	6
5	Tables and figures	7
5.1	Figures 1–2: aggregate puzzle and local competition	7
5.2	Figures 3–4: competition relation and model logic	8
5.3	Tables 1–2: loan and county summary statistics	8
5.4	Tables 3–4: baseline and within-originator results	9
5.5	Figure 5 and Tables 5–6: HUD-rule identification	10
5.6	Figure 6 and Tables 7–8: marginal borrowers and high competition	11
5.7	Tables 9–10: denial rates and delinquency consequences	12
6	Short summary	12
7	Conclusion	13
7.1	Core conclusion	13
7.2	Economic intuition	13
7.3	Takeaway	13

1 Big picture

- **Question.** Did competition among residential mortgage-backed security (RMBS) sponsors help create non-standard mortgage contracts before the financial crisis?
- **Core mechanism.** Securitization sponsors compete for investors and loan pools. Because financial products are hard to patent, sponsors can escape competition by differentiating the underlying loans, making products harder for rivals to imitate and evaluate.
- **Main empirical setting.** The paper links hand-collected RMBS sponsor information from prospectuses to over 18 million securitized loans from ABSNet over 1997–2008.
- **Product differentiation.** A loan is differentiated if it contains at least one non-standard feature: hybrid ARM, balloon, interest-only, negative amortization, pay teaser, or option ARM.
- **Competition measure.** Sponsor competition is measured by a county-year local sponsor Herfindahl–Hirschman Index (HHI) based on securitized loan balances. Lower HHI means more competition.
- **Puzzling aggregate facts.** Non-standard mortgage features move with private securitization activity, and differentiation rises first in Prime MBS collateral before spreading to non-prime collateral.
- **Baseline result.** Greater local sponsor competition is associated with higher loan-level product differentiation. A one-standard-deviation decrease in local sponsor HHI raises the probability of differentiated loan features by about 1.2–2.5 percentage points.
- **Identification.** The paper uses the 2000 HUD affordable-housing rule for GSE purchases as a regulatory shock that affected private sponsor competition differently across counties.
- **Within-originator design.** Results survive comparisons of loans from the same originator, in the same state or CBSA-year, but exposed to different local sponsor competition.
- **Prime then non-prime.** Differentiation initially rises in higher-quality Prime MBS collateral after competition increases; at very high competition, differentiation shifts toward marginal borrowers.
- **Crisis relevance.** Differentiated prime loans in the early boom period later generated unusually large delinquency amounts, suggesting that product differentiation helped obscure risk before the crisis.
- **Takeaway.** Securitization can create a feedback loop: competition among sellers of the derived security changes the design of the underlying asset.

2 Data and measurement

2.1 Loan-level data

- The core loan data come from Lewtan’s ABSNet and contain more than 18 million private-label securitized mortgage loans from 1997–2008.
- The data record loan features, property type, loan purpose, loan size, geography, MBS classification, and delinquency outcomes.
- MBS categories include Prime, Subprime, Alt-A, and other classifications.
- The paper focuses on securitized loans because the mechanism is about the feedback from RMBS sponsor competition to collateral design.

Interpretation: ABSNet provides the loan-level granularity needed to connect local competition among sponsors to the exact features of mortgages securitized into RMBS pools.

2.2 Product differentiation measure

A mortgage is treated as differentiated when it includes any non-standard feature:

$$\text{Differentiation}_{i,c,t} = 1\{\text{Hybrid ARM, Balloon, IO, NegAm, Pay Teaser, or Option ARM}\}.$$

- The measure is motivated by the idea that differentiated financial products have limited historical performance data.
- Lack of historical data makes default probabilities harder to assess and products harder to copy.
- Differentiation is measured both as a binary indicator and as intensity, the number of differentiated features in the same loan.

Interpretation: The measure captures product-design complexity in the underlying loan, not just complexity in the RMBS tranche structure.

2.3 Sponsor data and parent identification

- RMBS deals are often issued through special purpose vehicles.
- The authors hand-collect prospectus information to identify the sponsor’s parent financial institution.
- Sponsors report geographic diversification of loan pools, which motivates measuring competition at a local level.

Interpretation: The hand-collected sponsor link is central. Without parent identification, sponsor competition would be difficult to distinguish from originator competition or borrower demand.

2.4 Local sponsor competition

For county c and year t , the paper constructs local sponsor concentration as

$$HHI_{c,t} = \sum_s \left(\frac{Balance_{s,c,t}}{\sum_{s'} Balance_{s',c,t}} \right)^2.$$

- s indexes private-label RMBS sponsors.
- Lower HHI means more competition.
- The baseline variable is lagged local sponsor HHI, $LocalSponsorHHI_{c,t-1}$.
- The measure is computed at the county-year level using securitized loan balances.

Interpretation: Because HHI is a concentration index, negative coefficients on $LocalSponsorHHI$ mean that more competition increases differentiation.

2.5 County-level controls

The county-year controls include:

- population;
- employment;
- total personal income;
- loan demand, measured using HMDA mortgage applications relative to originations.

Interpretation: These variables address local demand and economic conditions that might jointly affect mortgage features and securitization competition.

3 Theory and empirical specifications

3.1 Theory: product differentiation as competition escape

The paper builds on Thakor’s idea that differentiated financial products are harder to evaluate and imitate because they lack long historical data. Sponsors competing in RMBS markets can therefore differentiate loan collateral to protect their positions.

The model has three dates:

- $t = 0$: the originator chooses whether to originate a standard or new mortgage in consultation with the sponsor.
- $t = 1$: the originator hands the loan to the sponsor, who creates tranches; a competing sponsor may enter.
- $t = 2$: investors are paid, or not paid, and the originator receives fee $f > 0$.

Interpretation: The timing creates a link between sponsor competition and mortgage design before the loan is securitized.

3.2 Borrower types and competition thresholds

The model distinguishes:

- **access borrowers:** regular borrowers with access to standard mortgage credit;
- **marginal borrowers:** borrowers who may not qualify for, or may not want, standard mortgages.

Let θ be the probability of competitive entry by another sponsor. The model predicts thresholds θ^* and θ^{**} :

$$\theta < \theta^* \Rightarrow \text{standard mortgages,}$$

$$\theta^* \leq \theta < \theta^{**} \Rightarrow \text{new mortgages offered to access borrowers,}$$

$$\theta \geq \theta^{**} \Rightarrow \text{new mortgages offered to both access and marginal borrowers.}$$

Interpretation: Competition first induces differentiation in higher-quality collateral. Only when competition is very high does it expand toward marginal borrowers.

3.3 Baseline loan-level specification

The baseline linear probability model is

$$Differentiation_{i,c,t} = \alpha + \beta LocalSponsorHHI_{c,t-1} + X'_{c,t-1}\gamma + \delta_{sponsor} + \delta_{state \times year} + \varepsilon_{i,c,t}. \quad (1)$$

- i indexes loans, c counties, and t years.
- $X_{c,t-1}$ contains local controls.
- Sponsor fixed effects absorb persistent sponsor risk-taking or strategy.
- State-by-year fixed effects absorb time-varying state regulatory and macro shocks.

Interpretation: Because lower HHI means more competition, the paper expects $\beta < 0$.

3.4 Within-originator specification

The stronger originator design adds fixed effects such as

$$\delta_{originator \times state \times year} \quad \text{or} \quad \delta_{originator \times CBSA \times year}.$$

Interpretation: This compares loans from the same originator in the same region-year, but exposed to different sponsor competition. It helps separate sponsor demand from originator-level borrower demand or risk appetite.

3.5 HUD-rule dynamic difference-in-differences

The regulatory strategy estimates

$$y_{c,t} = \alpha + \sum_{\tau} \beta_{\tau} (PrimeMarket_c \times \mathbf{1}\{t = \tau\}) + PrimeMarket_c + \sum_{\tau} \delta_{\tau} + \delta_c + X'_{c,t} \gamma + \varepsilon_{c,t}. \quad (2)$$

- $PrimeMarket_c$ identifies counties less affected by the GSE affordable-housing purchase shock.
- The 2000 HUD rule increased GSE purchases from underserved areas, reducing private-label sponsor competition more in lower-income areas.
- The treated prime markets are higher-income counties where private-label sponsors faced more intensified competition after the rule.

Interpretation: The rule helps isolate supply-side changes in sponsor competition that are less likely to be driven by borrower demand for non-standard loans.

3.6 Marginal borrower spline specification

To test whether high competition pushes differentiation toward marginal borrowers, the paper uses a spline around $SponsorHHI = 0.1$:

$$Y_{i,c,t} = \alpha + \beta_1 (SponsorHHI_{c,t-1}^{low} \times MarginalB_i) + \beta_2 (SponsorHHI_{c,t-1}^{high} \times MarginalB_i) \quad (3)$$

$$+ \beta_3 SponsorHHI_{c,t-1}^{low} + \beta_4 SponsorHHI_{c,t-1}^{high} + X'_{c,t-1} \gamma + FE + \varepsilon_{i,c,t}. \quad (4)$$

Interpretation: The sign and slope changes show whether marginal borrowers receive more differentiated products precisely at very high competition levels.

3.7 Delinquency analysis

The delinquency analysis compares the dollar value of delinquent loans by:

- securitization cohort;
- MBS class: Prime, Subprime, Alt-A;
- standard versus differentiated mortgage features.

Interpretation: This links the pre-crisis product-design pattern to ex post risk. The key question is whether sponsor-induced differentiated products subsequently produced large losses.

4 Identification and empirical design

4.1 What identifies the baseline relation

- The key variation is county-year sponsor competition interacted with loan-level mortgage features.
- Sponsor fixed effects absorb permanent sponsor differences.
- State-year fixed effects absorb broad regulatory and housing-market shocks.
- County controls and loan demand proxies address local borrower demand.

Interpretation: The baseline estimates document a strong conditional association between sponsor competition and loan design.

4.2 Why within-originator tests matter

- Originators, not sponsors, offer loan contracts directly to borrowers.
- The paper therefore compares loans from the same originator in the same state-year or CBSA-year.
- This rules out many originator-level stories, such as one lender becoming more aggressive everywhere.

Interpretation: If sponsor competition still matters within originator-region-year cells, the mechanism is more likely to run through sponsors' demand for differentiated collateral.

4.3 Why the HUD rule is useful

- The 2000 HUD rule raised affordable housing goals for GSEs.
- GSEs increased purchases in underserved areas, reducing private-label securitizer competition there.
- Higher-income prime markets were less crowded out, so private-label sponsors competed more intensely there.
- The paper uses this variation to trace changes in sponsor competition and then loan differentiation.

Interpretation: This regulatory shock provides more credible causal variation than simply comparing high- and low-HHI counties.

4.4 Remaining limitations

- The shock is not randomized at the loan level.
- Product differentiation may reflect multiple parties in the securitization chain.
- The data observe securitized loans, not all originated loans.
- Non-standard features may be attractive to borrowers for reasons not fully captured by controls.

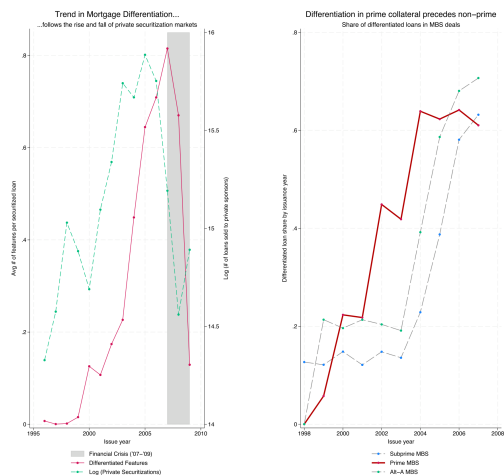
Interpretation: The paper's case is strongest because different empirical designs tell the same story: sponsor competition predicts where and when non-standard mortgage features emerge.

5 Tables and figures

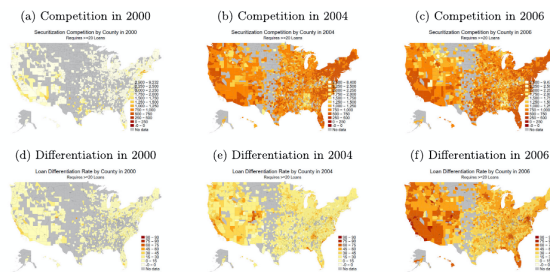
This section preserves the original paper’s visuals and places adjacent figures/tables together where possible to save space.

5.1 Figures 1–2: aggregate puzzle and local competition

(a) Why is sponsor activity connected to loan differentiation? (b) Why does prime differentiation precede that of non-prime MBS in the early boom period?



(a) Fig. 1: non-standard features, securitization, and Prime/non-prime timing.



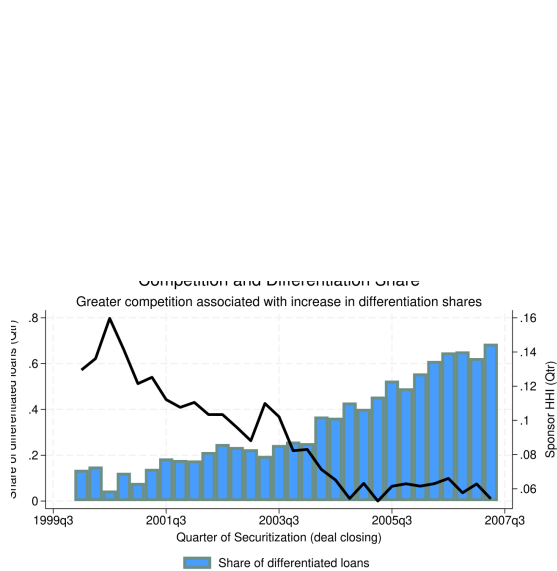
(b) Fig. 2: geography of sponsor competition and loan differentiation.

Read:

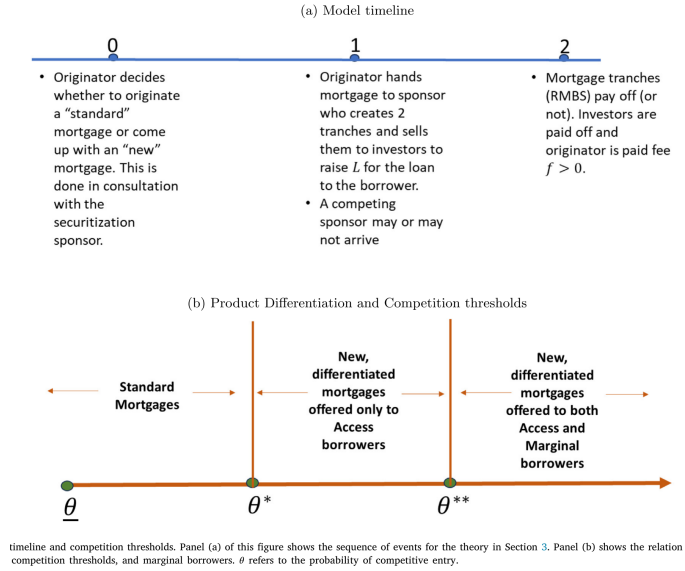
- Figure 1 shows that non-standard loan features rise and fall with private securitization activity.
- The right panel of Figure 1 shows differentiation rising first in Prime MBS collateral before rising in Subprime and Alt-A collateral.
- Figure 2 shows substantial geographic dispersion in sponsor competition and loan differentiation in 2000, 2004, and 2006.

Economic message: The paper begins from two facts that are hard to explain with borrower demand alone: securitization activity is tightly connected to loan differentiation, and the earliest differentiation appears in Prime collateral.

5.2 Figures 3–4: competition relation and model logic



(a) Fig. 3: lower sponsor HHI and higher differentiation.



(b) Fig. 4: model timeline and competition thresholds.

Read:

- Figure 3 shows sponsor HHI falling while the share of differentiated loans rises.
- Figure 4 shows the model sequence: origination choice, sponsor sale of tranches, possible entry, and payoff.
- The threshold diagram predicts standard mortgages at low competition, new mortgages for access borrowers at intermediate competition, and new mortgages for marginal borrowers at very high competition.

Economic message: The theory directly explains the empirical sequence: competition first produces differentiation in safer collateral, then expands toward marginal borrowers when competitive pressure becomes extreme.

5.3 Tables 1–2: loan and county summary statistics

Table 1
Loan characteristics.

(a) Statistics by property type

	Standard loans		Differentiated loans		All loans	
	N	Avg. size	N	Avg. size	N	Avg. size
Single family	7,086,275	\$199,788	5,517,863	\$283,221	12,604,138	\$236,313
PUD	926,719	\$232,257	1,259,843	\$284,065	2,186,562	\$262,107
Other	1,751,375	\$168,460	1,508,562	\$222,910	3,259,937	\$193,657

(b) Statistics by loan purpose

	Standard loans		Differentiated loans		All loans	
	N	Avg. size	N	Avg. size	N	Avg. size
Purchase	3,504,491	\$191,961	3,862,903	\$258,930	7,367,394	\$227,074
Cash Out Refi	3,209,151	\$194,675	2,348,908	\$292,096	5,558,059	\$255,846
Refinancing	1,686,349	\$262,250	1,275,100	\$349,602	2,961,449	\$299,861
Other	1,364,378	\$136,556	799,357	\$156,151	2,163,735	\$143,795

(c) Statistics by type of MBS

	Standard loans		Differentiated loans		All loans	
	N	Avg. size	N	Avg. size	N	Avg. size
Prime	1,551,076	\$395,703	1,812,687	\$419,067	3,363,763	\$410,071
Subprime	5,725,372	\$151,536	3,671,932	\$182,481	9,397,304	\$163,627
Alt-A	1,573,369	\$201,826	2,076,092	\$290,271	3,649,461	\$252,140
Other	914,352	\$138,990	725,557	\$259,592	1,640,109	\$192,342

This table describes loan characteristics of 18 million securitized loans over the period 1997–2008. Across all panels, summary statistics are shown separately for Standard and Differentiated loans, where a loan is defined as “Differentiated” if it takes any of the following features: Negative Amortization, Interest Only, Hybrid ARM, Option ARM, Balloon Payment or Pay Teaser. Standard loans are the complementary set of non-differentiated loans. N refers to the number of loans in each category and Avg. size refers to the mean of the original loan balance. In Panel (a), we provide summary statistics broken out by the property type. Note, PUD refers to “Planned Unit Development”. Panel (b) provides summary statistics by loan purpose, as defined in ABSNet. Finally, in Panel (c) we provide summary statistics by the type of MBS, which refers to the MBS category of the securitized deal in ABSNet.

(a) Table 1: loan characteristics.

Table 2
County characteristics.

(a) All local markets

	Averages across counties					
	N	Mean	σ	p50	p25	p75
Number of loans (#)	7720	2341	8974	366	98	1388
Number of differentiated loans (#)	7720	1074	3336	56	9	397
Local Sponsor HHI	7896	0.178	0.133	0.129	0.081	0.208
Loan Demand	7711	2.11	0.32	2.09	1.88	2.32
Population (#)	7471	345,109	393,038	165,516	97,727	380,949
Total Personal Income (\$ m)	7471	\$12.3	\$22.7	\$5.3	\$2.8	\$12.8
Employment (#)	7471	208,244	369,878	95,415	50,827	223,022

(b) Prime and non-Prime markets

	PrimeMarket			nonPrimeMarket		
	N	Mean	p50	N	Mean	p50
Number of loans (#)	4936	2744	381	2784	1627	350
Number of differentiated loans (#)	4936	1366	97	2784	723	54
Local Sponsor HHI	4923	0.177	0.129	2773	0.179	0.129
Loan Demand	4927	2.17	2.15	2784	2.02	1.99
Population (#)	4795	302,810	181,752	3076	259,636	155,842
Total Personal Income (\$ m)	4795	\$13.5	\$5.3	3076	\$10.1	\$5.2
Employment (#)	4795	242,552	105,672	3076	146,769	65,445

This table describes characteristics of local markets where sponsors compete for sourcing loans from originators. Summary statistics are presented at the county × year level over the period 1997–2008. Panel (a) summarizes data for all counties, whereas Panel (b) describes characteristics for the Trained (PrimeMarket) and Control (nonPrimeMarket) counties. If the percentage of census tract in a given county defined as Underlender Area as per the HUD regulation are in the 25th percentile or below, then that county is classified as a PrimeMarket. This definition captures higher-income areas. Additional variable definitions can be found in Internet Appendix Table IA.B.2.

(b) Table 2: county characteristics.

Read:

- Table 1 shows more than 18 million securitized loans, split into standard and differentiated loans.
- Differentiated loans are common across property types, loan purposes, and MBS categories.
- Prime differentiated loans are larger on average than many other loan groups.
- Table 2 shows strong dispersion across county-year markets in loan counts, differentiated loan counts, local sponsor HHI, and demand variables.

Economic message: The summary statistics justify the local-market approach: sponsor competition and differentiated lending vary substantially across counties and time.

5.4 Tables 3–4: baseline and within-originator results

Table 3
Higher sponsor competition associated with greater loan differentiation.

	Differentiation indicator			
	(1)	(2)	(3)	(4)
<i>LocalSponsorHHI_{c,j-1}</i>	-0.253*** (0.077)	-0.133** (0.054)	-0.133** (0.054)	-0.132*** (0.037)
<i>Population_{c,j-1}</i>		-0.131*** (0.017)	-0.106*** (0.017)	-0.062*** (0.010)
<i>Employment_{c,j-1}</i>		-0.035*** (0.009)	-0.034*** (0.009)	0.015** (0.006)
<i>Income_{c,j-1}</i>		0.167*** (0.014)	0.143*** (0.015)	0.042*** (0.007)
<i>LoanDemand_{c,j-1}</i>			-0.051*** (0.010)	-0.076*** (0.008)
Sponsor FE	Yes	Yes	Yes	Yes
State × Year FE	Yes	Yes	Yes	–
CBSA × Year FE	–	–	–	Yes
Observations	18,020,433	17,462,253	17,462,253	17,434,827
R ²	0.205	0.208	0.208	0.212

This table reports the results of linear probability panel regressions investigating:

$$1_{\text{Differentiation}_{i,j,t}} = \alpha + \beta_1(\text{LocalSponsorHHI}_{c,j-1}) + X_{c,j-1} + \delta_{\text{Sponsor}} + \delta_{\text{State} \times \text{Year}} + \epsilon$$

We estimate regressions over the database of securitized mortgages over the period 1997–2008. The dependent variable is an indicator that captures the incidence of differentiation in a securitized loan. Differentiation for individual loans is calculated based on indicators for any of the following features: Hybrid ARM, Balloon, Interest Only, Negative Amortization, Pay Teaser, Option ARM. The main independent variable refers to lagged Sponsor HHI (Herfindahl–Hirschman Index) based on loan balances defined at the county-year level. Columns 2–4 include county-level, time-varying controls. Columns 3–4 include loan demand which is measured as the lagged ratio of HMDA applications to HMDA originations in a given county and year. The fixed effects included in each regression are denoted below the regression. Standard errors are clustered at the State × Year level. Detailed variable definitions can be found in Internet Appendix Table IA.B.2.

(a) Table 3: baseline competition and differentiation.

Table 4
Greater sponsor competition within the same originator leads to higher loan differentiation.

	Differentiation indicator			
	(1)	(2)	(3)	(4)
<i>LocalSponsorHHI_{c,j-1}</i>	-0.226*** (0.065)	-0.120*** (0.044)	-0.121*** (0.044)	-0.096*** (0.027)
<i>Population_{c,j-1}</i>		-0.109*** (0.013)	-0.090*** (0.013)	-0.050*** (0.007)
<i>Employment_{c,j-1}</i>		-0.029*** (0.007)	-0.029*** (0.007)	0.011** (0.004)
<i>Income_{c,j-1}</i>		0.138*** (0.011)	0.121*** (0.012)	0.036*** (0.006)
<i>LoanDemand_{c,j-1}</i>			-0.038*** (0.007)	-0.061*** (0.007)
Sponsor FE	Yes	Yes	Yes	Yes
Originator × State × Year FE	Yes	Yes	Yes	–
Originator × CBSA × Year FE	–	–	–	Yes
Observations	17,782,788	17,234,366	17,234,366	17,137,086
R ²	0.358	0.360	0.360	0.371

This table reports the results of linear probability panel regressions investigating:

$$1_{\text{Differentiation}_{i,j,t}} = \alpha + \beta_1(\text{LocalSponsorHHI}_{c,j-1}) + X_{c,j-1} + \delta_{\text{Sponsor}} + \delta_{\text{Originator} \times \text{State} \times \text{Year}} + \epsilon$$

We estimate regressions over the database of securitized mortgages over the period 1997–2008. The dependent variable is an indicator that captures the incidence of differentiation in a securitized loan. Differentiation for individual loans is calculated based on indicators for any of the following features: Hybrid ARM, Balloon, Interest Only, Negative Amortization, Pay Teaser, Option ARM. Importantly, the empirical strategy includes Originator × State × Year fixed effects, so the identifying variation compares the same originator in the same state at the same point in time, but exposed to differing levels of sponsor competition. The main independent variable refers to lagged Sponsor HHI (Herfindahl–Hirschman Index) based on loan balances defined at the county-year level. Columns 2–4 include county-level, time-varying controls. Columns 3–4 include loan demand which is measured as the lagged ratio of HMDA applications to HMDA originations in a given county and year. The fixed effects included in each regression are denoted below the regression. Standard errors are clustered at the State × Year level. All variables are defined in Internet Appendix

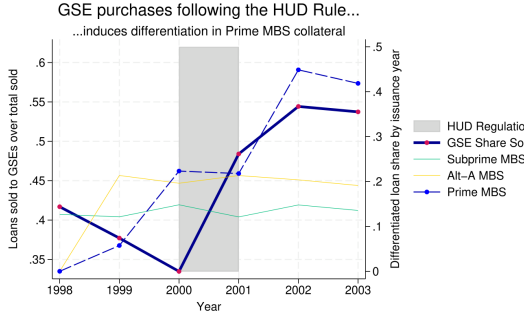
(b) Table 4: within-originator competition and differentiation.

Read:

- Table 3 estimates negative coefficients on local sponsor HHI across specifications, so greater competition raises the probability of differentiation.
- The preferred estimate remains significant after adding county-level controls, loan demand, sponsor fixed effects, and state-year or CBSA-year fixed effects.
- Table 4 repeats the test within originator-by-state-year and originator-by-CBSA-year cells.
- Even within the same originator and local time period, lower sponsor HHI predicts more non-standard loan features.

Economic message: These tables provide the paper’s main evidence that sponsors, not just local borrowers or originators, shaped mortgage product design.

5.5 Figure 5 and Tables 5–6: HUD-rule identification



(a) Fig. 5: GSE purchases after the HUD rule.

Table 5
Change in sponsor competition in prime markets after the HUD rule.

	Double difference estimates Local Sponsor HHI		
	(1)	(2)	(3)
PrimeMarket × 1997	-0.011 (0.009)	-0.011 (0.009)	-0.014 (0.010)
PrimeMarket × 1998	-0.003 (0.006)	-0.003 (0.006)	-0.006 (0.005)
PrimeMarket × 2000	-0.022*** (0.006)	-0.022*** (0.006)	-0.026*** (0.006)
PrimeMarket × 2001	-0.018*** (0.006)	-0.018*** (0.006)	-0.019*** (0.006)
PrimeMarket × 2002	-0.013*** (0.005)	-0.013*** (0.005)	-0.015*** (0.005)
PrimeMarket × 2003	-0.007 (0.005)	-0.007 (0.005)	-0.011** (0.005)
PrimeMarket × 2004	-0.008* (0.005)	-0.008* (0.004)	-0.012*** (0.005)
PrimeMarket × 2005	-0.006 (0.005)	-0.006 (0.005)	-0.010* (0.005)
PrimeMarket × 2006	-0.007 (0.005)	-0.007 (0.005)	-0.010** (0.005)
PrimeMarket × 2007	-0.006 (0.005)	-0.006 (0.005)	-0.012** (0.005)
PrimeMarket × 2008	-0.004 (0.010)	-0.004 (0.010)	-0.008 (0.010)
Control for loan demand?	No	Yes	Yes
Control for time-varying County Char.?	No	No	Yes
County FE	Yes	Yes	Yes
State × Year FE	Yes	Yes	Yes
Observations	7655	7646	7192
R ²	0.823	0.823	0.741

This table reports the results of dynamic differences-in-differences panel regressions investigating:

(b) Table 5: sponsor competition after HUD rule.

Table 6
Differentiation induced in prime markets by the HUD rule of 2000.

	Double difference estimates Loan differentiation		
	(1)	(2)	(3)
PrimeMarket × 1997	0.002 (0.011)	0.002 (0.011)	0.005 (0.013)
PrimeMarket × 1998	-0.004 (0.009)	-0.004 (0.009)	-0.002 (0.012)
PrimeMarket × 2000	0.014 (0.010)	0.014 (0.010)	0.010 (0.011)
PrimeMarket × 2001	0.030** (0.012)	0.029** (0.012)	0.022** (0.011)
PrimeMarket × 2002	0.041*** (0.014)	0.044*** (0.014)	0.032*** (0.012)
PrimeMarket × 2003	0.035*** (0.008)	0.036*** (0.008)	0.021** (0.010)
PrimeMarket × 2004	0.047*** (0.007)	0.048*** (0.007)	0.038*** (0.009)
PrimeMarket × 2005	0.041*** (0.007)	0.042*** (0.007)	0.021** (0.009)
PrimeMarket × 2006	0.032*** (0.008)	0.033*** (0.007)	0.009 (0.010)
PrimeMarket × 2007	0.022*** (0.008)	0.022*** (0.008)	-0.001 (0.011)
PrimeMarket × 2008	0.001 (0.013)	0.004 (0.013)	-0.021 (0.014)
Control for loan demand?	No	Yes	Yes
Control for time-varying County Char.?	No	No	Yes
Sponsor FE	Yes	Yes	Yes
County FE	Yes	Yes	Yes
State × Year FE	Yes	Yes	Yes
Observations	18,022,263	17,721,204	17,462,253
R ²	0.212	0.212	0.211

This table reports the results of dynamic differences-in-differences panel regressions investigating:

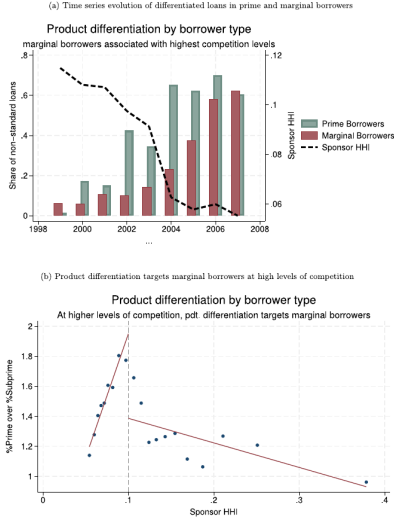
(c) Table 6: differentiation after HUD rule.

Read:

- Figure 5 shows GSE purchases rising after the 2000 HUD rule, especially during the rule window.
- Table 5 shows that local sponsor HHI falls in PrimeMarket counties after the rule, implying stronger private-label sponsor competition there.
- Table 6 shows loan differentiation rising in PrimeMarket counties after the same shock, with larger coefficients during 2001–2004.

Economic message: The regulatory shock links competition and differentiation in sequence: HUD changes GSE behavior, private sponsor competition rises in prime markets, and differentiated features increase there.

5.6 Figure 6 and Tables 7–8: marginal borrowers and high competition



(a) Fig. 6: differentiation by borrower type and competition.

Table 7
Competition, product differentiation and marginal borrowers.

Spline regression	Product Differentiation Indicator			
	(1)	(2)	(3)	(4)
SponsorHHI ^{low} × Marginal Borrower	-5.949*** (0.743)	-5.870*** (0.733)	-5.866*** (0.735)	-6.112*** (0.457)
SponsorHHI ^{high} × Marginal Borrower	1.108*** (0.241)	1.176*** (0.214)	1.168*** (0.217)	1.333*** (0.165)
SponsorHHI ^{low}	3.800*** (0.795)	4.117*** (0.683)	4.064*** (0.681)	4.282*** (0.428)
SponsorHHI ^{high}	-1.082*** (0.254)	-1.045*** (0.210)	-1.033*** (0.213)	-1.128*** (0.159)
Marginal Borrower	0.288*** (0.062)	0.288*** (0.061)	0.288*** (0.061)	0.311*** (0.038)
Controls	No	Yes	Yes	Yes
Sponsor FE	Yes	Yes	Yes	Yes
Originator × State × Year FE	Yes	Yes	Yes	Yes
Originator × CBSA × Year FE	–	–	–	Yes
Observations	16,319,290	15,817,053	15,817,053	15,727,555
R ²	0.363	0.364	0.364	0.374

This table reports the results of a spline regression investigating:

$$1_{\text{Differentiation}_{i,t,j}} = \alpha + \beta_1(\text{SponsorHHI}_{i,t,j-1}^{\text{low}} \times \text{MarginalB.}) + \beta_2(\text{SponsorHHI}_{i,t,j-1}^{\text{high}} \times \text{MarginalB.}) + \beta_3(\text{SponsorHHI}_{i,t,j-1}^{\text{low}}) + \beta_4(\text{SponsorHHI}_{i,t,j-1}^{\text{high}}) + X_{i,t,j-1} + \delta_{\text{Sponsor}} + \delta_{\text{Originator} \times \text{State} \times \text{Year}} + \epsilon$$

We estimate regressions over the database of securitized mortgages over the period 1997–2008. The dependent variable is an indicator that captures the incidence of product differentiation in a securitized loan. Differentiation for individual loans is calculated based on indicators for any of the following features: Hybrid ARM, Balloon, Interest Only, Negative Amortization, Pay Teaser, Option ARM. Importantly, the empirical strategy includes Originator × State × Year fixed effects, so the identifying variation compares the same originator in the same state at the same point in time, but exposed to differing levels of sponsor competition. The two main variables SponsorHHI^{low} and SponsorHHI^{high} are splines of SponsorHHI, defined using the cutoff point SponsorHHI = 0.1 motivated by Panel (b) of Fig. 6. MarginalBorrower is an indicator for borrowers classified as Subprime and Alt-A. Columns 2–4 include county-level, time-varying controls. Columns 3–4 include loan demand which is measured as the lagged ratio of HMDA applications to HMDA originations in a given county and year. The fixed effects included in each regression are denoted below the regression. Standard errors are clustered at the State × Year level. All variables are defined in Internet Appendix Table I.A.B.2.

Competition, product differentiation intensity and marginal borrowers.

Spline regression	Degree of Product Differentiation			
	(1)	(2)	(3)	(4)
SponsorHHI ^{low} × Marginal Borrower	-5.902*** (1.836)	-5.835*** (1.800)	-5.827*** (1.805)	-6.330*** (1.059)
SponsorHHI ^{high} × Marginal Borrower	2.377*** (0.453)	2.545*** (0.410)	2.529*** (0.415)	2.828*** (0.317)
SponsorHHI ^{low}	2.956 (1.848)	3.681** (1.623)	3.568** (1.621)	4.534*** (0.829)
SponsorHHI ^{high}	-2.196*** (0.459)	-2.131*** (0.385)	-2.106*** (0.390)	-2.257*** (0.309)
Marginal Borrower	0.043 (0.165)	0.050 (0.163)	0.050 (0.164)	0.097 (0.095)
Controls	No	Yes	Yes	Yes
Sponsor FE	Yes	Yes	Yes	Yes
Originator × State × Year FE	Yes	Yes	Yes	–
Originator × CBSA × Year FE	–	–	–	Yes
Observations	16,319,290	15,817,053	15,817,053	15,727,555
R ²	0.459	0.460	0.460	0.469

This table reports the results of a spline regression investigating:

$$y_{i,t,j} = \alpha + \beta_1(\text{SponsorHHI}_{i,t,j-1}^{\text{low}} \times \text{MarginalB.}) + \beta_2(\text{SponsorHHI}_{i,t,j-1}^{\text{high}} \times \text{MarginalB.}) + \beta_3(\text{SponsorHHI}_{i,t,j-1}^{\text{low}}) + \beta_4(\text{SponsorHHI}_{i,t,j-1}^{\text{high}}) + X_{i,t,j-1} + \delta_{\text{Sponsor}} + \delta_{\text{Originator} \times \text{State} \times \text{Year}} + \epsilon$$

We estimate regressions over the database of securitized mortgages over the period 1997–2008. The dependent variable y_i is a count of the number of differentiated features introduced in the same securitized loan. Differentiation for individual loans is based on any of the following features: Hybrid ARM, Balloon, Interest Only, Negative Amortization, Pay Teaser, Option ARM. Importantly, the empirical strategy includes Originator × State × Year fixed effects, so the identifying variation compares the same originator in the same state at the same point in time, but exposed to differing levels of sponsor competition. The two main variables SponsorHHI^{low} and SponsorHHI^{high} are splines of SponsorHHI, defined using the cutoff point SponsorHHI = 0.1 motivated by Panel (b) of Fig. 6. MarginalBorrower is an indicator for borrowers classified as Subprime and Alt-A. Columns 2–4 include county-level, time-varying controls. Columns 3–4 include loan demand which is measured as the lagged ratio of HMDA applications to HMDA originations in a given county and year. The fixed effects included in each regression are denoted below the regression. Standard errors are clustered at the State × Year level. All variables are defined in Internet Appendix Table I.A.B.2.

(b) Tables 7–8: marginal borrowers and differentiated features.

Read:

- Figure 6 shows differentiation among marginal borrowers rising in the years just before the crisis.
- At lower competition levels, prime borrowers have more differentiated loans than marginal borrowers.
- At very high competition, the pattern reverses: differentiation targets marginal borrowers.
- Table 7 confirms this with the product-differentiation indicator.
- Table 8 shows similar patterns for the degree of differentiation.

Economic message: The evidence matches the model’s second prediction. Sponsors initially differentiate high-quality loans; at intense competition levels, they push differentiated products toward marginal borrowers.

5.7 Tables 9–10: denial rates and delinquency consequences

Table 9

Mortgage denial rates are lowest when high competition targets marginal borrowers.

Spline regression	Denial rates (County-level)		
	(1)	(2)	(3)
SponsorHHI ^{low} × MarginalB.Share	1.159*** (0.320)	0.721** (0.307)	0.755*** (0.288)
SponsorHHI ^{high} × MarginalB.Share	−0.020 (0.021)	0.048* (0.029)	0.096*** (0.033)
MarginalBorrowerShare	−0.098*** (0.030)	−0.075*** (0.028)	−0.080*** (0.026)
SponsorHHI ^{low}	−1.513*** (0.292)	−1.067*** (0.282)	−1.087*** (0.264)
SponsorHHI ^{high}	0.018 (0.019)	−0.038 (0.026)	−0.083*** (0.031)
Controls	No	Yes	Yes
County FE	Yes	Yes	Yes
State × Year FE	Yes	Yes	Yes
Observations	7653	7653	7255
R ²	0.844	0.878	0.884

This table reports the results of a spline regression investigating:

$$y_{c,t} = \alpha + \beta_1(\text{SponsorHHI}_{t,j-1}^{\text{low}} \times \text{MarginalB.Share}) + \beta_2(\text{SponsorHHI}_{t,j-1}^{\text{high}} \times \text{MarginalB.Share}) + \beta_3(\text{SponsorHHI}_{t,j-1}^{\text{low}}) + \beta_4(\text{SponsorHHI}_{t,j-1}^{\text{high}}) + X_{c,t-1} + \delta_{\text{County}} + \delta_{\text{State} \times \text{Year}} + \epsilon$$

We estimate regressions over the database of securitized mortgages over the period 1997–2008. The dependent variable $y_{c,t}$ is the ratio of number of mortgage applications denied a given County-Year over the total applications in a given County-Year. We perform this regression at the county level because we do not have a direct mapping of ABSNet with HMDA at the loan level. The two main variables SponsorHHI^{low} and SponsorHHI^{high} are splines of SponsorHHI, defined using the cutoff point SponsorHHI = 0.1 motivated by Panel (b) of Fig. 6. MarginalBorrowerShare is computed as the share of borrowers classified as Subprime and Alt-A in the ABSNet database. Column (2) includes county loan demand as a control and column (3) includes time-varying county-level characteristics as controls. The fixed effects included in each regression are denoted below the regression. Standard errors are clustered at the State × Year level. All variables are defined in Internet Appendix Table IA.B.2.

Table 10

Delinquencies by product differentiation and securitization cohort.

(Values in \$ bn)	Subprime		Alt-A		Prime	
	Standard (1)	Differentiated (2)	Standard (3)	Differentiated (4)	Standard (5)	Differentiated (6)
<i>Early boom period</i>						
2000	5.64	0.19	0.31	0.03	0.53	0.66
2001	5.46	0.37	3.13	2.98	5.69	1.47
2002	19.39	1.18	2.96	3.68	14.93	19.31
2003	36.8	4.86	7.54	2.23	29.04	26.85
2004	56.0	17.90	10.02	7.34	18.32	42.90
Sub-total	123.29	24.51	23.97	16.26	68.52	91.20
<i>Pre-crisis period</i>						
2005	78.09	58.31	11.85	42.99	15.88	33.47
2006	69.48	132.14	17.50	87.78	11.74	37.48
2007	25.46	75.72	14.06	69.51	9.70	27.92
2008	0.58	0.60	0.02	0.28	0.55	0.62
Sub-total	173.61	266.77	43.43	200.56	37.9	99.49
Total	296.91	291.28	67.40	216.82	106.40	190.68

This table presents the economic significance of delinquencies in the collateral of Subprime, Alt-A and Prime MBS. Note that the years do not refer to the year of delinquency, rather they refer to the year of securitization (deal closing). Values are in billions of dollars based on loan balances at origination. The delinquency amount within each category is further broken down into standard and differentiated collateral. A loan is defined as “Differentiated” if it takes any of the following features: Negative Amortization, Interest Only, Hybrid ARM, Option ARM, Balloon Payment or Pay Teaser. All variables are defined in Internet Appendix Table IA.B.2.

(a) Table 9: denial rates and high competition.

(b) Table 10: delinquencies by cohort and product type.

Read:

- Table 9 shows that mortgage denial rates are lowest when high competition targets marginal borrowers.
- Table 10 reports delinquency dollar amounts by cohort, MBS type, and whether loans are standard or differentiated.
- In the early boom period, differentiated Prime MBS collateral eventually accounts for more delinquency dollars than standard Prime collateral.
- In the later pre-crisis period, differentiated loans in non-prime segments generate much larger delinquency amounts.

Economic message: Differentiation is not merely cosmetic. The product-design changes are tied to expanded access and later credit deterioration.

6 Short summary

- The paper studies whether RMBS sponsor competition caused mortgage-level product differentiation before the financial crisis.
- It uses 18 million securitized loans from ABSNet and hand-collected sponsor-parent data from RMBS prospectuses.
- Product differentiation means a mortgage contains features such as hybrid ARM, balloon, interest-only, negative amortization, pay teaser, or option ARM.
- The main competition measure is local sponsor HHI at the county-year level.
- Lower HHI, meaning more competition, predicts higher probability of differentiated mortgage features.
- The result survives sponsor fixed effects, state-year fixed effects, county controls, loan demand controls, and within-originator comparisons.

- The identification strategy uses the 2000 HUD rule, which changed GSE purchases and indirectly shifted private-label sponsor competition.
- After the HUD shock, PrimeMarket counties see more private sponsor competition and more loan-level differentiation.
- The theory predicts thresholds: standard mortgages at low competition, differentiated mortgages for access borrowers at intermediate competition, and differentiated mortgages for marginal borrowers at very high competition.
- The empirical evidence supports this threshold logic.
- Differentiated Prime collateral in the early boom period later generated unusually large delinquency amounts.
- The broader lesson is that competition in the market for derived securities can change the underlying asset itself.

7 Conclusion

7.1 Core conclusion

Haslag, Srinivasan and Thakor show that securitization sponsors were not passive purchasers of pre-existing mortgage products. Their competition influenced the design of the loans that became RMBS collateral. Greater sponsor competition induced originators to issue more non-standard loans, first in Prime collateral and later among marginal borrowers.

7.2 Economic intuition

Financial products are difficult to patent. When RMBS sponsors face stronger competition, they can differentiate their collateral to make products harder to imitate and evaluate. This differentiation initially targets access borrowers, where quality is higher and uncertainty is lower. As competition becomes more intense, sponsors expand differentiation to marginal borrowers, lowering denial rates but increasing future credit risk.

7.3 Takeaway

The paper provides evidence of a feedback loop from securities markets to real financial contracts. Competition in securitization does not only affect how loans are pooled or tranced; it can alter the mortgage contracts households receive. This helps explain the rise of non-standard mortgages before the crisis and why differentiated products contributed to subsequent delinquencies.